

# Birmingham Parking Dataset - EDA Report

## Overview

### Overview

#### Dataset statistics

|                               |         |
|-------------------------------|---------|
| Number of variables           | 4       |
| Number of observations        | 35717   |
| Missing cells                 | 0       |
| Missing cells (%)             | 0.0%    |
| Duplicate rows                | 160     |
| Duplicate rows (%)            | 0.4%    |
| Total size in memory          | 4.9 MiB |
| Average record size in memory | 144.9 B |

#### Variable types

|             |   |
|-------------|---|
| Categorical | 1 |
| Numeric     | 2 |
| DateTime    | 1 |

### Alerts 4

#### Alerts

|                                                                                                         |                  |
|---------------------------------------------------------------------------------------------------------|------------------|
| Dataset has 160 (0.4%) <a href="#">duplicate rows</a>                                                   | Duplicates       |
| <a href="#">Capacity</a> is highly overall correlated with <a href="#">occupancy</a> and 1 other fields | High correlation |
| <a href="#">occupancy</a> is highly overall correlated with <a href="#">Capacity</a>                    | High correlation |
| <a href="#">SystemCodeNumber</a> is highly overall correlated with <a href="#">Capacity</a>             | High correlation |

### Reproduction

#### Reproduction

|                        |                                         |
|------------------------|-----------------------------------------|
| Analysis started       | 2026-10-03 06:32:02.388945              |
| Analysis finished      | 2026-10-03 06:32:04.110115              |
| Duration               | 1.72 second                             |
| Software version       | <a href="#">ydata-profiling v4.18.4</a> |
| Download configuration | <a href="#">config.json</a>             |

# Variables

## SystemCodeNumber

Categorical

High correlation

|              |         |              |       |
|--------------|---------|--------------|-------|
| Distinct     | 30      | BHMBCCMK     | 1312  |
| Distinct (%) | 0.1%    | BHMBCCMK     | 1312  |
| Missing      | 0       | BHMBCCMK     | 1312  |
| Missing (%)  | 0.0%    | BHMBCCMK     | 1312  |
| Memory size  | 2.1 MiB | Other values | 29157 |

### Overview

#### Length

|               |           |
|---------------|-----------|
| Max length    | 16        |
| Median length | 11        |
| Mean length   | 11.877565 |
| Min length    | 8         |

#### Characters and Unicode

|                     |      |
|---------------------|------|
| Total characters    | 4242 |
| Distinct characters | 43   |
| Distinct categories | 1    |
| Distinct scripts    | 1    |
| Distinct blocks     | 1    |

The Unicode Standard assigns character properties to each code point, which can be used to analyse textual variables.

#### Unique

|            |      |   |
|------------|------|---|
| Unique     | 0    | ? |
| Unique (%) | 0.0% |   |

#### Sample

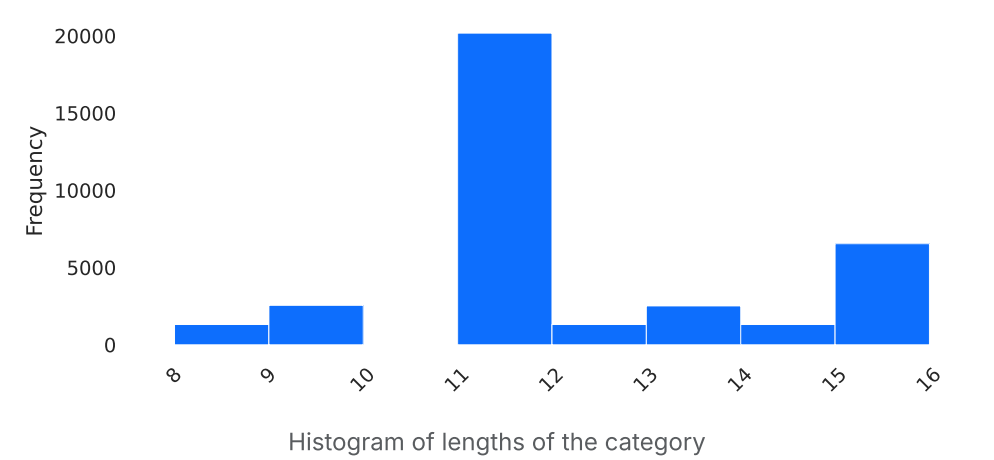
|         |          |
|---------|----------|
| 1st row | BHMBCCMK |
| 2nd row | BHMBCCMK |
| 3rd row | BHMBCCMK |
| 4th row | BHMBCCMK |
| 5th row | BHMBCCMK |

Categories

Common Values

| Value             | Count | Frequency (%) |
|-------------------|-------|---------------|
| BHMBCCMKT01       | 1312  | 3.7%          |
| BHMBCCTHL01       | 1312  | 3.7%          |
| BHMMBMMBX01       | 1312  | 3.7%          |
| BHMNCPHST01       | 1312  | 3.7%          |
| BHMEURBRD01       | 1312  | 3.7%          |
| Shopping          | 1312  | 3.7%          |
| Others-CCCPS105a  | 1312  | 3.7%          |
| Others-CCCPS119a  | 1312  | 3.7%          |
| Others-CCCPS135a  | 1312  | 3.7%          |
| Others-CCCPS202   | 1312  | 3.7%          |
| Other values (20) | 22597 | 63.3%         |

Length



Words

| Value             | Count | Frequency (%) |
|-------------------|-------|---------------|
| nia               | 2570  | 6.1%          |
| bhmbccmkt01       | 1312  | 3.1%          |
| bhmbcctl01        | 1312  | 3.1%          |
| bhmmbmmbx01       | 1312  | 3.1%          |
| bhmeurbrd01       | 1312  | 3.1%          |
| bhmncphst01       | 1312  | 3.1%          |
| others-cccps105a  | 1312  | 3.1%          |
| others-cccps119a  | 1312  | 3.1%          |
| others-cccps135a  | 1312  | 3.1%          |
| shopping          | 1312  | 3.1%          |
| Other values (24) | 27611 | 65.8%         |

Characters

Characters

Most occurring characters

| Value             | Count  | Frequency (%) |
|-------------------|--------|---------------|
| C                 | 50167  | 11.8%         |
| B                 | 40279  | 9.5%          |
| H                 | 26419  | 6.2%          |
| M                 | 25419  | 6.0%          |
| 1                 | 23065  | 5.4%          |
| O                 | 22795  | 5.4%          |
| S                 | 20517  | 4.8%          |
| P                 | 20368  | 4.8%          |
| N                 | 14993  | 3.5%          |
| R                 | 14840  | 3.5%          |
| Other values (33) | 165369 | 39.0%         |

Categories

Most occurring categories

| Value     | Count  | Frequency (%) |
|-----------|--------|---------------|
| (unknown) | 424231 | 100.0%        |

## Most frequent character per category

(unknown)

| Value             | Count  |             | Frequency (%) |
|-------------------|--------|-------------|---------------|
| C                 | 50167  | <div></div> | 11.8%         |
| B                 | 40279  | <div></div> | 9.5%          |
| H                 | 26419  | <div></div> | 6.2%          |
| M                 | 25419  | <div></div> | 6.0%          |
| 1                 | 23065  | <div></div> | 5.4%          |
| O                 | 22795  | <div></div> | 5.4%          |
| S                 | 20517  | <div></div> | 4.8%          |
| P                 | 20368  | <div></div> | 4.8%          |
| N                 | 14993  | <div></div> | 3.5%          |
| R                 | 14840  | <div></div> | 3.5%          |
| Other values (33) | 165369 | <div></div> | 39.0%         |

## Scripts

### Most occurring scripts

| Value     | Count  |             | Frequency (%) |
|-----------|--------|-------------|---------------|
| (unknown) | 424231 | <div></div> | 100.0%        |

## Most frequent character per script

(unknown)

| Value             | Count  |             | Frequency (%) |
|-------------------|--------|-------------|---------------|
| C                 | 50167  | <div></div> | 11.8%         |
| B                 | 40279  | <div></div> | 9.5%          |
| H                 | 26419  | <div></div> | 6.2%          |
| M                 | 25419  | <div></div> | 6.0%          |
| 1                 | 23065  | <div></div> | 5.4%          |
| O                 | 22795  | <div></div> | 5.4%          |
| S                 | 20517  | <div></div> | 4.8%          |
| P                 | 20368  | <div></div> | 4.8%          |
| N                 | 14993  | <div></div> | 3.5%          |
| R                 | 14840  | <div></div> | 3.5%          |
| Other values (33) | 165369 | <div></div> | 39.0%         |

Blocks

Most occurring blocks

| Value     | Count  | Frequency (%) |
|-----------|--------|---------------|
| (unknown) | 424231 | 100.0%        |

Most frequent character per block

(unknown)

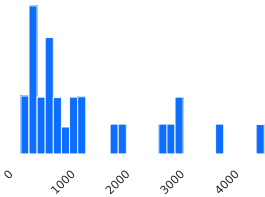
| Value             | Count  | Frequency (%) |
|-------------------|--------|---------------|
| C                 | 50167  | 11.8%         |
| B                 | 40279  | 9.5%          |
| H                 | 26419  | 6.2%          |
| M                 | 25419  | 6.0%          |
| 1                 | 23065  | 5.4%          |
| O                 | 22795  | 5.4%          |
| S                 | 20517  | 4.8%          |
| P                 | 20368  | 4.8%          |
| N                 | 14993  | 3.5%          |
| R                 | 14840  | 3.5%          |
| Other values (33) | 165369 | 39.0%         |

Capacity

Real number ( $\mathbb{R}$ )

High correlation

|              |           |              |           |
|--------------|-----------|--------------|-----------|
| Distinct     | 30        | Minimum      | 220       |
| Distinct (%) | 0.1%      | Maximum      | 4675      |
| Missing      | 0         | Zeros        | 0         |
| Missing (%)  | 0.0%      | Zeros (%)    | 0.0%      |
| Infinite     | 0         | Negative     | 0         |
| Infinite (%) | 0.0%      | Negative (%) | 0.0%      |
| Mean         | 1397.5501 | Memory size  | 279.2 KiB |



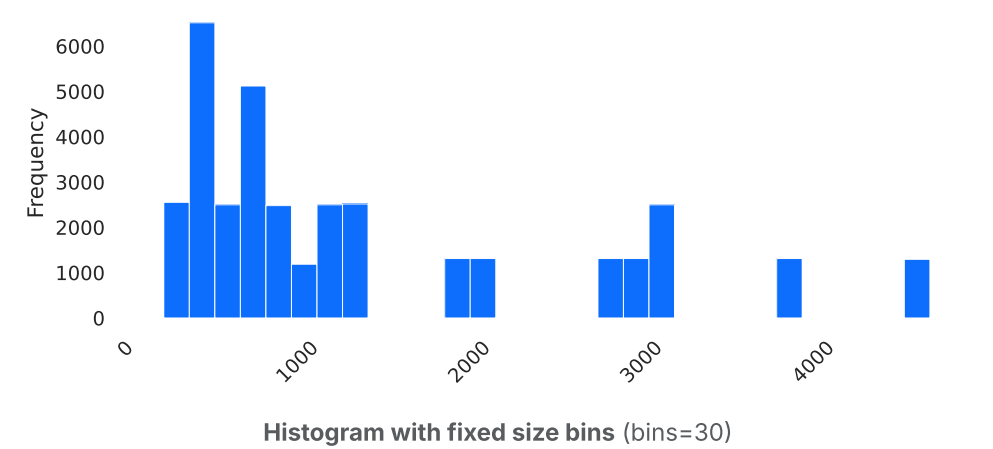
Statistics

Quantile statistics

Descriptive statistics

|                           |      |                                 |               |
|---------------------------|------|---------------------------------|---------------|
| Minimum                   | 220  | Standard deviation              | 1179.3268     |
| 5-th percentile           | 317  | Coefficient of variation (CV)   | 0.84385297    |
| Q1                        | 500  | Kurtosis                        | 0.53063676    |
| median                    | 849  | Mean                            | 1397.5501     |
| Q3                        | 2009 | Median Absolute Deviation (MAD) | 399           |
| 95-th percentile          | 3883 | Skewness                        | 1.2624422     |
| Maximum                   | 4675 | Sum                             | 49916298      |
| Range                     | 4455 | Variance                        | 1390811.8     |
| Interquartile range (IQR) | 1509 | Monotonicity                    | Not monotonic |

### Histogram



### Common values

| Value             | Count | Frequency (%) |
|-------------------|-------|---------------|
| 577               | 1312  | 3.7%          |
| 387               | 1312  | 3.7%          |
| 687               | 1312  | 3.7%          |
| 1200              | 1312  | 3.7%          |
| 470               | 1312  | 3.7%          |
| 1920              | 1312  | 3.7%          |
| 2009              | 1312  | 3.7%          |
| 2803              | 1312  | 3.7%          |
| 3883              | 1312  | 3.7%          |
| 2937              | 1312  | 3.7%          |
| Other values (20) | 22597 | 63.3%         |

Extreme values

Minimum 10 values

| Value | Count | Frequency (%)   |      |
|-------|-------|-----------------|------|
| 220   | 1276  | <div>3.6%</div> |      |
| 317   | 1276  | <div>3.6%</div> |      |
| 387   | 1312  | <div>3.7%</div> |      |
| 450   | 1291  | <div>3.6%</div> |      |
| 470   | 1312  | <div>3.7%</div> |      |
| 480   | 162   | <div></div>     | 0.5% |
| 485   | 1312  | <div>3.7%</div> |      |
| 496   | 88    | <div></div>     | 0.2% |
| 500   | 1038  | <div>2.9%</div> |      |
| 577   | 1312  | <div>3.7%</div> |      |

Maximum 10 values

| Value | Count | Frequency (%)   |  |
|-------|-------|-----------------|--|
| 4675  | 1294  | <div>3.6%</div> |  |
| 3883  | 1312  | <div>3.7%</div> |  |
| 3103  | 1312  | <div>3.7%</div> |  |
| 3053  | 1186  | <div>3.3%</div> |  |
| 2937  | 1312  | <div>3.7%</div> |  |
| 2803  | 1312  | <div>3.7%</div> |  |
| 2009  | 1312  | <div>3.7%</div> |  |
| 1920  | 1312  | <div>3.7%</div> |  |
| 1322  | 1312  | <div>3.7%</div> |  |
| 1268  | 1204  | <div>3.4%</div> |  |

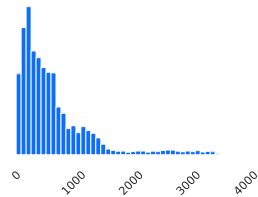
Occupancy

Real number ( $\mathbb{R}$ )

High correlation



|                     |           |                     |           |
|---------------------|-----------|---------------------|-----------|
| <b>Distinct</b>     | 3013      | <b>Minimum</b>      | -8        |
| <b>Distinct (%)</b> | 8.4%      | <b>Maximum</b>      | 4327      |
| <b>Missing</b>      | 0         | <b>Zeros</b>        | 17        |
| <b>Missing (%)</b>  | 0.0%      | <b>Zeros (%)</b>    | < 0.1%    |
| <b>Infinite</b>     | 0         | <b>Negative</b>     | 12        |
| <b>Infinite (%)</b> | 0.0%      | <b>Negative (%)</b> | < 0.1%    |
| <b>Mean</b>         | 642.22891 | <b>Memory size</b>  | 279.2 KiB |



### Statistics

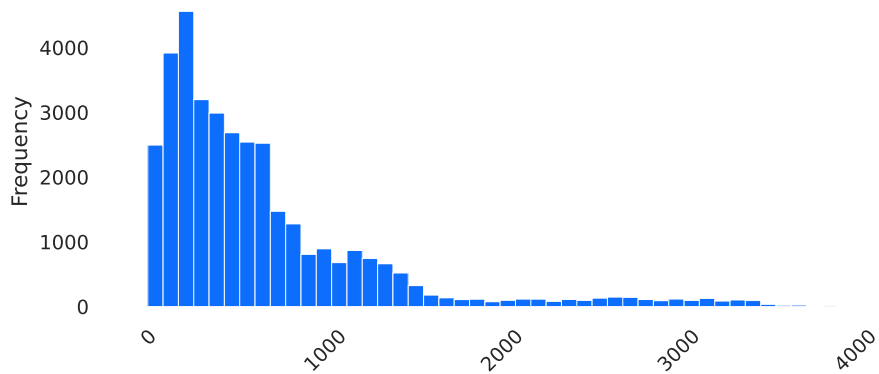
#### Quantile statistics

|                                  |      |
|----------------------------------|------|
| <b>Minimum</b>                   | -8   |
| <b>5-th percentile</b>           | 63   |
| <b>Q1</b>                        | 210  |
| <b>median</b>                    | 446  |
| <b>Q3</b>                        | 798  |
| <b>95-th percentile</b>          | 2140 |
| <b>Maximum</b>                   | 4327 |
| <b>Range</b>                     | 4335 |
| <b>Interquartile range (IQR)</b> | 588  |

#### Descriptive statistics

|                                        |               |
|----------------------------------------|---------------|
| <b>Standard deviation</b>              | 656.95553     |
| <b>Coefficient of variation (CV)</b>   | 1.0229305     |
| <b>Kurtosis</b>                        | 5.4533918     |
| <b>Mean</b>                            | 642.22891     |
| <b>Median Absolute Deviation (MAD)</b> | 261           |
| <b>Skewness</b>                        | 2.2042411     |
| <b>Sum</b>                             | 22938490      |
| <b>Variance</b>                        | 431590.57     |
| <b>Monotonicity</b>                    | Not monotonic |

### Histogram



Histogram with fixed size bins (bins=50)

Common values

| Value               | Count | Frequency (%) |
|---------------------|-------|---------------|
| 402                 | 92    | 0.3%          |
| 345                 | 86    | 0.2%          |
| 219                 | 74    | 0.2%          |
| 206                 | 73    | 0.2%          |
| 170                 | 73    | 0.2%          |
| 215                 | 73    | 0.2%          |
| 1010                | 71    | 0.2%          |
| 153                 | 70    | 0.2%          |
| 201                 | 68    | 0.2%          |
| 169                 | 67    | 0.2%          |
| Other values (3003) | 34970 | 97.9%         |

Extreme values

Minimum 10 values

| Value | Count | Frequency (%) |
|-------|-------|---------------|
| -8    | 1     | < 0.1%        |
| -4    | 1     | < 0.1%        |
| -3    | 3     | < 0.1%        |
| -2    | 2     | < 0.1%        |
| -1    | 5     | < 0.1%        |
| 0     | 17    | < 0.1%        |
| 1     | 15    | < 0.1%        |
| 2     | 6     | < 0.1%        |
| 3     | 8     | < 0.1%        |
| 4     | 8     | < 0.1%        |

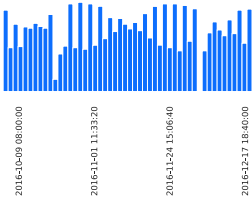
Maximum 10 values

| Value | Count | Frequency (%) |
|-------|-------|---------------|
| 4327  | 1     | < 0.1%        |
| 4289  | 1     | < 0.1%        |
| 4272  | 1     | < 0.1%        |
| 4270  | 1     | < 0.1%        |
| 4153  | 1     | < 0.1%        |
| 4139  | 1     | < 0.1%        |
| 4073  | 1     | < 0.1%        |
| 4070  | 1     | < 0.1%        |
| 4017  | 1     | < 0.1%        |
| 3983  | 1     | < 0.1%        |

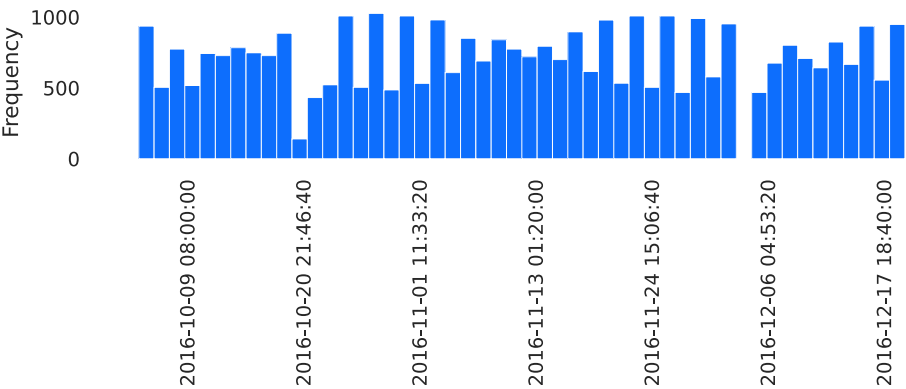
LastUpdated

Date

|              |           |                   |               |
|--------------|-----------|-------------------|---------------|
| Distinct     | 1784      | Minimum           | 2016-10-04 0  |
| Distinct (%) | 5.0%      | Maximum           | 2016-12-19 16 |
| Missing      | 0         | Invalid dates     | 0             |
| Missing (%)  | 0.0%      | Invalid dates (%) | 0.0%          |
| Memory size  | 279.2 KiB |                   |               |



Histogram

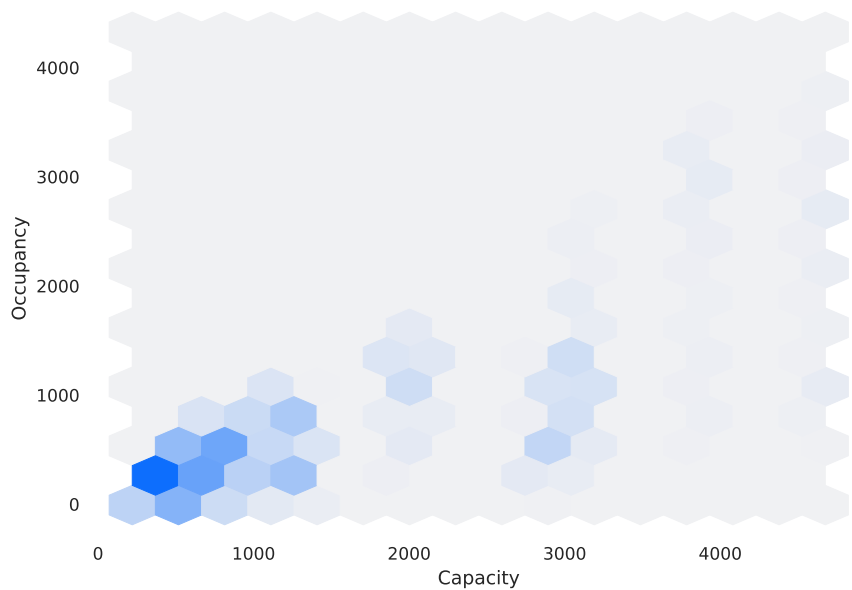


Histogram with fixed size bins (bins=50)

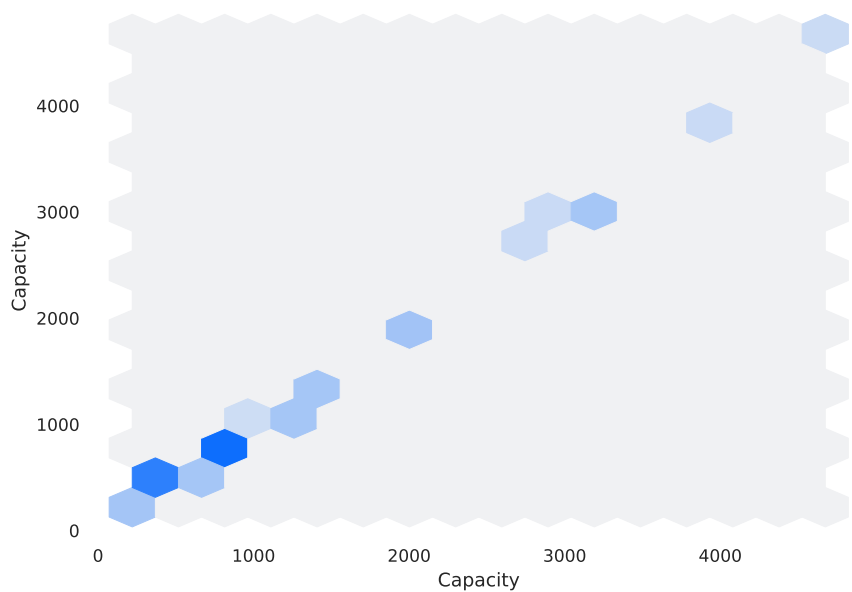
# Interactions

Capacity

Occupancy

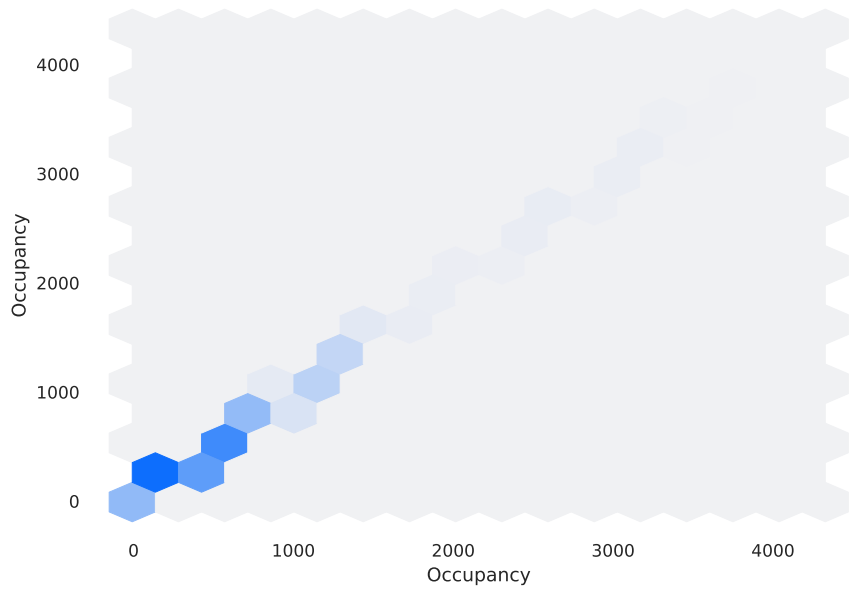


Capacity

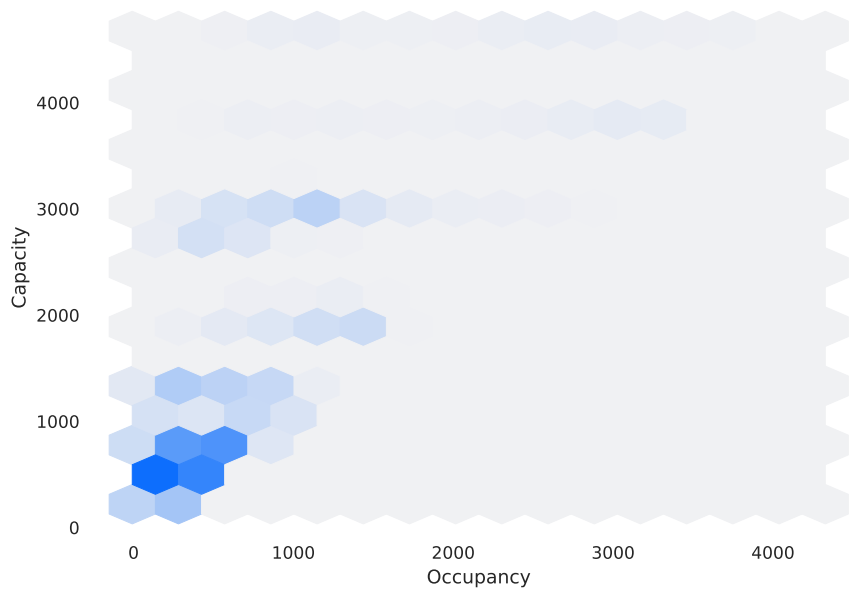


Occupancy

Occupancy



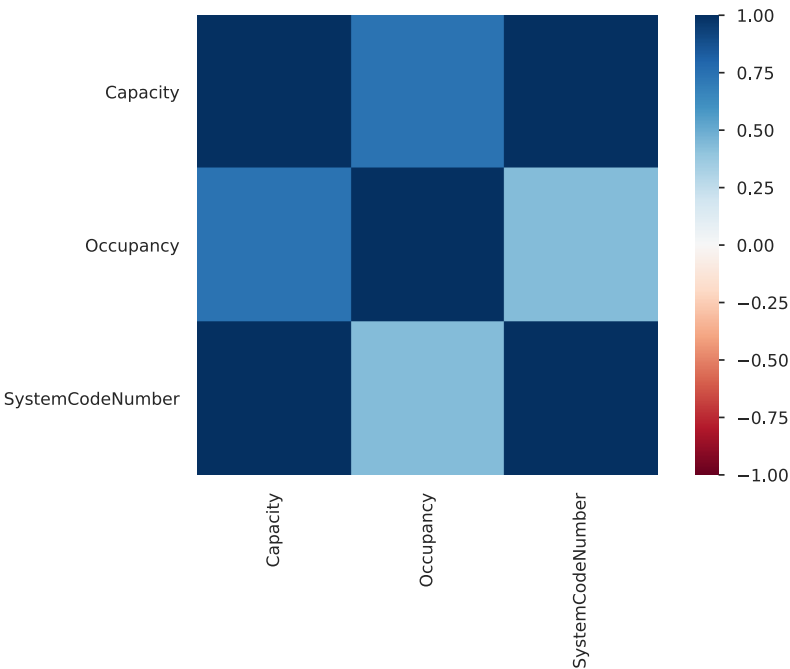
Capacity



# Correlations

Auto

Heatmap

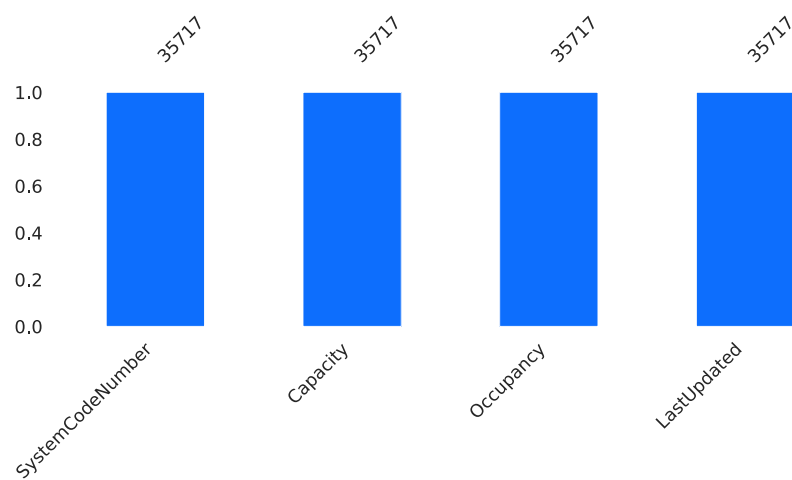


Table

|                  | Capacity | Occupancy | SystemCodeNumber |
|------------------|----------|-----------|------------------|
| Capacity         | 1.000    | 0.737     | 1.000            |
| Occupancy        | 0.737    | 1.000     | 0.437            |
| SystemCodeNumber | 1.000    | 0.437     | 1.000            |

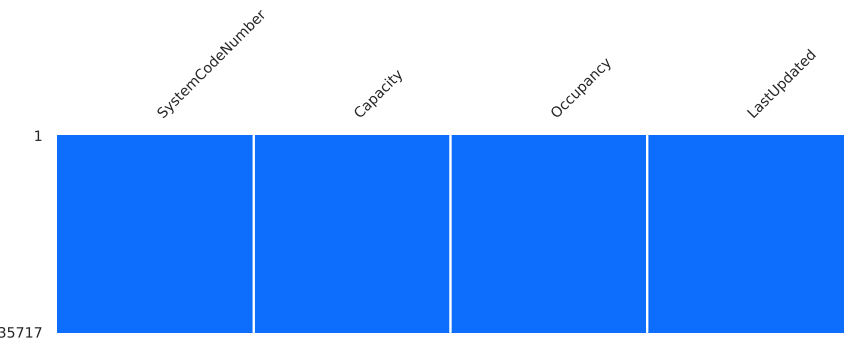
# Missing values

Count



A simple visualization of nullity by column.

Matrix



Nullity matrix is a data-dense display which lets you quickly visually pick out patterns in data completion.

# Sample

## First rows

|   | SystemCodeNumber | Capacity | Occupancy | LastUpdated         |
|---|------------------|----------|-----------|---------------------|
| 0 | BHMBCCMKT01      | 577      | 61        | 2016-10-04 07:59:42 |
| 1 | BHMBCCMKT01      | 577      | 64        | 2016-10-04 08:25:42 |
| 2 | BHMBCCMKT01      | 577      | 80        | 2016-10-04 08:59:42 |
| 3 | BHMBCCMKT01      | 577      | 107       | 2016-10-04 09:32:46 |
| 4 | BHMBCCMKT01      | 577      | 150       | 2016-10-04 09:59:48 |
| 5 | BHMBCCMKT01      | 577      | 177       | 2016-10-04 10:26:49 |
| 6 | BHMBCCMKT01      | 577      | 219       | 2016-10-04 10:59:48 |
| 7 | BHMBCCMKT01      | 577      | 247       | 2016-10-04 11:25:47 |
| 8 | BHMBCCMKT01      | 577      | 259       | 2016-10-04 11:59:44 |
| 9 | BHMBCCMKT01      | 577      | 266       | 2016-10-04 12:29:45 |

## Last rows

|       | SystemCodeNumber | Capacity | Occupancy | LastUpdated         |
|-------|------------------|----------|-----------|---------------------|
| 35707 | Shopping         | 1920     | 1424      | 2016-12-19 12:03:32 |
| 35708 | Shopping         | 1920     | 1475      | 2016-12-19 12:30:32 |
| 35709 | Shopping         | 1920     | 1510      | 2016-12-19 13:03:30 |
| 35710 | Shopping         | 1920     | 1510      | 2016-12-19 13:30:35 |
| 35711 | Shopping         | 1920     | 1521      | 2016-12-19 14:03:31 |
| 35712 | Shopping         | 1920     | 1517      | 2016-12-19 14:30:33 |
| 35713 | Shopping         | 1920     | 1487      | 2016-12-19 15:03:34 |
| 35714 | Shopping         | 1920     | 1432      | 2016-12-19 15:29:33 |
| 35715 | Shopping         | 1920     | 1321      | 2016-12-19 16:03:35 |
| 35716 | Shopping         | 1920     | 1180      | 2016-12-19 16:30:35 |



# Duplicate rows

## Most frequently occurring

|     | SystemCodeNumber | Capacity | Occupancy | LastUpdated            | #<br>duplicates |
|-----|------------------|----------|-----------|------------------------|-----------------|
| 61  | BHMNCPNHS01      | 500      | 90        | 2016-11-28<br>07:55:24 | 5               |
| 67  | BHMNCPNST01      | 485      | 193       | 2016-11-28<br>07:42:26 | 5               |
| 77  | BHMNCPPLS01      | 450      | 9         | 2016-10-30<br>08:25:53 | 5               |
| 104 | BHMNCPRAN01      | 600      | 197       | 2016-11-28<br>07:55:24 | 5               |
| 55  | BHMNCPLDH01      | 720      | 328       | 2016-11-28<br>07:55:24 | 4               |
| 58  | BHMNCPLDH01      | 720      | 360       | 2016-12-06<br>07:56:10 | 4               |
| 73  | BHMNCPNST01      | 485      | 309       | 2016-10-30<br>07:52:54 | 4               |
| 81  | BHMNCPPLS01      | 450      | 11        | 2016-11-13<br>09:34:15 | 4               |
| 87  | BHMNCPPLS01      | 450      | 18        | 2016-11-27<br>08:52:22 | 4               |
| 1   | BHMBCCMKT01      | 577      | 43        | 2016-10-30<br>07:59:55 | 3               |